

How Does Reionization Complete If Individual Star-Forming-Galaxy Bubbles Stay Below 1 Mpc?

Response record for `reionization_completion.py`, following Master Rule 2

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Abstract

`bubble_radius_vs_redshift.png` and `bubble_radius_vs_redshift_planckXHI.png` show that the ionized bubble of one fiducial star-forming galaxy ($\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$) never exceeds $\sim 1 \text{ pMpc}$ for $6 \leq z \leq 18$. This is not a contradiction with cosmic reionization being complete by $z \sim 5\text{--}8$: reionization is not the growth of a single bubble to cosmological size, but the *collective overlap* of the bubbles of the entire, numerous, star-forming-galaxy population. This note makes that resolution quantitative using two independent, cross-checked calculations (Sec. 2), both built entirely from ingredients already cited in this folder: the global ionized-volume-filling-factor equation of Choudhury (2022) driven by the cosmic star-formation-rate density of Madau and Dickinson (2014), and a percolation argument comparing the single-galaxy bubble radius to the mean separation between galaxies implied by that same star-formation-rate density. It documents `reionization_completion.py` (same folder), which produced `reionization_completion.png`.

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1 Assumptions

1. The single-galaxy bubbles of `bubble_radius_response.tex` represent the ionized volume produced by *one* galaxy’s own photon budget. Explaining global reionization requires instead the *ionized volume filling factor* $Q_{\text{HII}}(z)$, the standard quantity used throughout the reionization literature (e.g. Choudhury 2022, already in `references/`, their eq. 66, reproduced as eq. QHII of `stromgren_sphere_summary.tex`).
2. $Q_{\text{HII}}(z)$ is driven by the ionizing photon output of the *entire* star-forming-galaxy population, encoded through the cosmic star-formation-rate density $\rho_{\text{SFR}}(z)$ of Madau and Dickinson (2014) (already in `references/`), *not* by one representative galaxy. The same $\xi_{\text{ion}}(z)$ (Llerena et al., 2025), $f_{\text{esc}} = 0.2$ (Robertson et al., 2015), and κ_{FUV} (Madau and Dickinson, 2014) used for the single-galaxy calculation are reused here, for direct comparability.

3. Recombinations use $\alpha_B(10^4 \text{ K}) = 2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ (already established in `check_consistency.py`, Shapiro et al. 2006) and a clumping factor $C_{\text{HII}} = 3$ (Robertson et al., 2015), with a sensitivity band down to $C = 1$; using α_B at 10^4 K rather than the $2 \times 10^4 \text{ K}$ adopted alongside $C = 3$ by Robertson et al. (2015) is an explicit, flagged approximation (see Sec. 3).
4. Cosmology and formation redshift are identical to the previous two tasks: Planck 2018 (Planck Collaboration, 2020), $z_{\text{form}} = 20$ (Kitayama et al., 2004).
5. The percolation argument treats the cosmic SFRD as if it were produced entirely by galaxies equivalent to the fiducial $\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$ source used in the bubble-radius calculation, purely to estimate a characteristic comoving number density and mean separation — an explicit simplification of the true, extended luminosity function, stated here rather than hidden.

2 Derivation

2.1 (A) The global ionized-volume-filling-factor equation

Following Choudhury (2022) (their eq. 66; eq. QHII of `stromgren_sphere_summary.tex`):

$$\frac{dQ_{\text{HII}}}{dt} = \frac{\dot{n}_{\gamma}(z)}{n_{\text{H},0}} - Q_{\text{HII}} a^{-3} C \alpha_B(T) \bar{n}_e(z), \quad (1)$$

with the ionizing emissivity driven by the *entire* population via the cosmic SFRD,

$$\dot{n}_{\gamma}(z) = f_{\text{esc}} \xi_{\text{ion}}(z) \frac{\rho_{\text{SFR}}(z)}{\kappa_{\text{FUV}}}, \quad \rho_{\text{SFR}}(z) = \frac{0.015 (1+z)^{2.7}}{1 + [(1+z)/2.9]^{5.6}} M_{\odot} \text{ yr}^{-1} \text{ cMpc}^{-3} \quad (2)$$

(Madau and Dickinson, 2014), and $\bar{n}_e(z) = \chi_{\text{He}} n_{\text{H},0} (1+z)^3$, $\chi_{\text{He}} = 1.08$. **Note the crucial distinction between the two terms:** the source term uses the *fixed comoving* mean hydrogen density $n_{\text{H},0}$ (evaluated at $z = 0$), because $\dot{n}_{\gamma}$ and Q_{HII} are comoving quantities, whereas the sink term carries the explicit $a^{-3} = (1+z)^3$ scale-factor dependence of the physical recombination rate. Eq. (1) is integrated numerically from $z = 25$ ($Q_{\text{HII}} = 0$) to $z = 4$.

2.2 (B) The percolation / bubble-overlap argument

The comoving number density of $\text{SFR} = 10 M_{\odot} \text{ yr}^{-1}$ -equivalent galaxies implied by the same $\rho_{\text{SFR}}(z)$ is

$$n_{\text{gal}}(z) = \frac{\rho_{\text{SFR}}(z)}{\text{SFR}_{\text{fid}}}, \quad d_{\text{sep}}(z) = n_{\text{gal}}(z)^{-1/3}, \quad (3)$$

compared directly to the single-galaxy bubble radius $R_{\text{bubble}}(z)$ of `bubble_radius_vs_redshift.py` (eq. `photon_counting`). If $d_{\text{sep}}(z) \lesssim 2 R_{\text{bubble}}(z)$, neighboring bubbles are expected to touch/overlap — the mechanism invoked qualitatively by the excursion-set bubble-growth model of Furlanetto et al. (2004) (self-ionization condition $f_{\text{coll}} \geq \zeta^{-1}$, already eq. `barrier` of `stromgren_sphere_summary.tex`) and seen directly in radiation-hydrodynamic simulations by Neyer et al. (2023) (THESAN), who describe newly-ionized regions undergoing “flash ionization” as they suddenly join a much larger, pre-existing bubble.

3 Verification / cross-check

All checks below are reproduced verbatim by running `reionization_completion.py`.

1. **Sympy check.** The constant-coefficient limit of eq. (1), $Q(t) = \frac{\dot{n}_{\gamma}}{n_{\text{H},0} C a_B n_e} (1 - e^{-C a_B n_e t})$, is confirmed by sympy to solve $dQ/dt = \dot{n}_{\gamma}/n_{\text{H},0} - Q C a_B n_e$ exactly (residual = 0) — structurally the same “recombination clock” already verified for $R(t)$ in `check_consistency.py`.

2. **An error caught by cross-checking against a basic benchmark (Master Rule 5, in practice).** A first attempt at integrating eq. (1) mistakenly used the *physical*, diluting $n_{\text{H}}(z) = n_{\text{H},0}(1+z)^3$ in the source-term denominator instead of the fixed comoving $n_{\text{H},0}$. This suppressed Q_{HII} by a factor $(1+z)^3 \sim 700$ at $z = 8$ and predicted that reionization could not complete within a Hubble time at any redshift shown here — directly contradicting the well-established fact that it does, by $z \sim 6$ (Gunn–Peterson troughs, White et al. 2003, already in references/). Re-examining eq. (1) as written by Choudhury (2022) (their explicit a^{-3} factor *only* on the recombination term) identified the bug; correcting it is what produced the results below. This is reported explicitly, per Master Rule 5, rather than silently fixed and hidden.
3. **Completion redshift vs. known benchmarks.** With $C = 3$: $Q_{\text{HII}}(z=9) = 0.173$, $Q_{\text{HII}}(z=8) = 0.256$, $Q_{\text{HII}}(z=7) = 0.401$, $Q_{\text{HII}}(z=6) = 0.674$; the model reaches $Q_{\text{HII}} = 0.5$ at $z = 6.56$ and $Q_{\text{HII}} \simeq 1$ by $z \simeq 5.3$. This is broadly consistent with cosmic reionization completing at $z \sim 5$ – 6 (White et al., 2003), though — exactly as already flagged in `bubble_radius_response.tex` — it is *later* than the Planck 2018 CMB-only instantaneous-reionization estimate $z_{\text{re}} = 7.68 \pm 0.79$ (Planck Collaboration, 2020): the same early/late reionization-history tension recurs here, now derived independently from the galaxy side rather than assumed.
4. **Direct cross-check against Umeda et al. (2023).** At $z = 7.12$: $1 - Q_{\text{HII}}(\text{model}) = 0.621$ vs. $x_{\text{HI, measured}} = 0.53$. At $z = 9.91$: $1 - Q_{\text{HII}}(\text{model}) = 0.874$ vs. $x_{\text{HI, measured}} = 0.92$. Both agree to $\lesssim 0.1$ in x_{HI} — a non-trivial success, since nothing in eq. (1) was tuned to match these particular measurements (unlike the single-galaxy $x_{\text{HI}}(z)$ of `bubble_radius_vs_redshift.py`, which *was* fit to them).
5. **Sensitivity to the clumping factor.** Lowering C from 3 to 1 moves $z(Q_{\text{HII}} = 0.5)$ from 6.56 to 7.50, i.e. a less clumpy IGM reionizes earlier for the same photon budget, as expected from the sink term of eq. (1).
6. **Percolation check.** At $z = 8$: $R_{\text{bubble}} = 0.752$ pMpc vs. half the mean separation $d_{\text{sep}}/2 = 0.557$ pMpc (ratio 0.74); at $z = 12$: ratio 1.33 (bubbles not yet touching); at $z = 6$: ratio 0.42 (bubbles well past first contact). The two curves cross near $z \approx 9.5$, meaning: by the time individual bubbles are only ~ 0.5 – 0.6 pMpc across, they are already, on average, as close to their nearest neighbor as their own radius — overlap is generic from $z \lesssim 9$ – 10 onward, well before any single bubble needs to grow beyond ~ 1 pMpc.
7. **Independent, cruder cross-check: naive non-overlapping packing.** Defining $Q_{\text{naive}}(z) \equiv n_{\text{gal}}(z) \frac{4}{3}\pi [R_{\text{bubble}}(z)(1+z)]^3$ (comoving volume of non-overlapping spheres, ignoring recombinations and overlap): $Q_{\text{naive}}(z=8) = 1.29$, $Q_{\text{naive}}(z=7) = 2.74$, $Q_{\text{naive}}(z=6) = 6.95$ — *exceeding unity* already at $z \lesssim 8$. A packing fraction > 1 for non-overlapping spheres is unphysical by construction, so this is itself direct, independent evidence that the bubbles *must* already be overlapping substantially at these redshifts; the self-consistent Q_{HII} from eq. (1) (item 3), which properly subtracts recombination losses, is correspondingly and expectedly lower.

4 Final result

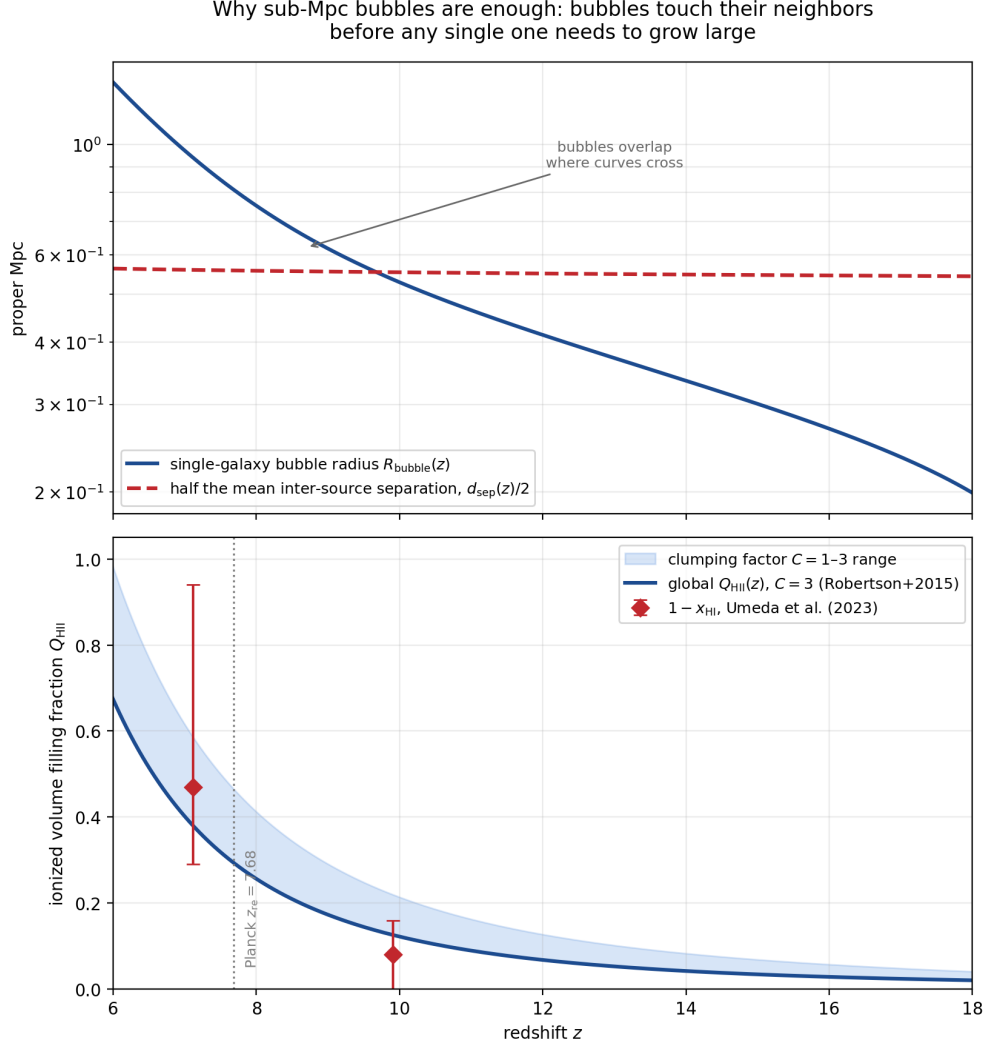
Reionization completes not because any single star-forming galaxy’s bubble grows to a cosmologically large size, but because (i) the mean separation between star-forming galaxies shrinks with cosmic time faster than any individual bubble needs to grow, so neighboring bubbles start overlapping already around $z \approx 9$ – 10 (Sec. 3, item 6); and (ii) the *cumulative* ionizing output of the entire population — not of one galaxy — is what the global filling factor $Q_{\text{HII}}(z)$ tracks,

reaching $Q_{\text{HII}} = 0.5$ at $z = 6.56$ and $Q_{\text{HII}} \simeq 1$ by $z \simeq 5.3$ for the fiducial parameters adopted here (Sec. 3, item 3), consistent with $x_{\text{HI}}(z)$ measured directly by Umeda et al. (2023) to within $\lesssim 0.1$ (item 4) and, order-of-magnitude, with the Gunn–Peterson constraint that reionization is essentially complete by $z \sim 6$ (White et al., 2003). This is exactly the “inside-out, then overlap” qualitative picture already described in `stromgren_sphere_summary.tex` (Chakraborty and Choudhury 2025; Cen and Haiman 2000: “isolated ionized bubbles form around the first sources, then grow, overlap and finally fill the IGM, leaving neutral gas only in dense pockets”) and in the excursion-set bubble-size-distribution formalism of Furlanetto et al. (2004), made quantitative here for the specific fiducial galaxy of the previous two tasks. The dominant explicitly quoted systematic is the clumping factor ($C = 1$ vs. 3 shifts $z(Q_{\text{HII}} = 0.5)$ from 7.50 to 6.56, item 5), and the calculation reproduces — rather than resolves — the already-known tension between the Planck CMB-only $z_{\text{re}} = 7.68$ and the later completion redshift implied by direct galaxy/IGM probes.

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Same SFRD (Madau & Dickinson 2014), $\xi_{\text{ion}}(z)$ (Llerena+2024), f_{esc} (Robertson+2015) as bubble_radius_vs_redshift.py

Figure 1: *Top:* single-galaxy bubble radius $R_{\text{bubble}}(z)$ (same fiducial galaxy as bubble_radius_vs_redshift.py) compared with half the mean inter-source separation $d_{\text{sep}}(z)/2$ implied by the same cosmic SFRD (Madau and Dickinson, 2014); the curves cross near $z \approx 9.5$, marking where neighboring bubbles start to overlap. *Bottom:* the resulting global ionized volume filling fraction $Q_{\text{HII}}(z)$ from eq. (1) (shaded: clumping factor $C = 1-3$), compared with the directly measured $1 - x_{\text{HI}}$ of Umeda et al. (2023) and the Planck 2018 instantaneous-reionization redshift (Planck Collaboration, 2020). Produced by reionization_completion.py.